CURRICULUM VITAE

Donghwan Lee

February 2026

EDUCATION

2017 - Present	Korea University, Ph.D. Candidate Major: Cyberwarfare (Thesis: Optimization of Block Withholding Attacks on Blockchain Networks) Advisor: Wonjun Lee, Ph.D.
2006 - 2008	Korea University, M.Sc. Major: Computer Science and Communications Engineering Advisor: Wonjun Lee, Ph.D.
2000 - 2006	Korea University, B.A. Major: Industrial System & Information Engineering Minor: Computer Science

PROFESSIONAL EXPERIENCE

2025 – 2026	**L2 Researcher**, Tokamak Network
2008 - 2025	**Senior Researcher**, Agency for Defense Development
2005 - 2006	**Associate Researcher**, NHN Games
2004 - 2005	**Internship — IT Sales & Marketing,** Sun Microsystems Korea
2003 - 2006	**President/Founder,** KuNiX (One of KU's earliest hacking clubs)

TECHNICAL SKILLS

Programming Languages	Python, C/C++, Java, Javascript/Typescript, Swift, Solidity
Frameworks & Tools	Tensorflow, PyTorch, Scikit-Learn

Data & Analytics	Strong background in optimization and quantitative analytics + R, Pandas, Numpy, MATLAB, SQL
Security & Networking	IDA Pro, Metasploit, Snort/Zeek/Suricata, Wireshark
Domain Expertise	Game-Theoretic, Statistical & Economic Analysis;
	Blockchain Security & Smart Contract Analysis;
	Cryptography and Secure Protocol Design;
	Threat Hunting & Offensive Security Operations;
	Machine Learning Model Development & Deployment

LANGUAGES

English	Professional Proficiency
Korean	Native
Japanese	Conversational

PUBLICATIONS

PEER-REVIEWED JOURNAL ARTICLES (INTERNATIONAL)

D. Lee and W. Lee, "Return to Dilemma: A Novel Block Withholding Attack on PoW-Based Blockchains," *IEEE Transactions on Computational Social Systems*, Early Access, 2025.

H. J. Yoon, A. R. Thukkaraju, S. Matsumoto, J. F. Ferrari, **D. Lee**, M. K. Ahn, "Subjective Bayesian Network-based Interdependent Mission Impact Assessment with Game-Theoretic Attack-Defense Interactions," *IEEE Internet of Things Journal*, 2024.

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D. Lee, D. Kim, C. Lee, M. K. Ahn, and W. Lee, "ICSTASY: An Integrated Cybersecurity Training System for Military Personnel," *IEEE Access*, vol. 10, pp. 62232–62246, 2022.

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D. Lee, D. Kim, M. K. Ahn, W. Jang, and W. Lee, "Cy-through: Toward a Cybersecurity Simulation for Supporting Live, Virtual, and Constructive Interoperability," *IEEE Access*, vol. 9, pp. 10041–10053, 2021.

S. Y. Enoch, Z. Huang, C. Y. Moon, **D. Lee**, M. K. Ahn and D. S. Kim, "HARMer: Cyber-Attacks Automation and Evaluation," *IEEE Access*, vol. 8, pp. 129397-129414, 2020.

D. Lee, J. Choi, and W. Lee. "OFSA: Optimum Frame-Slotted Aloha for RFID Tag Collision Arbitration," *KSII Transactions on Internet & Information Systems*, vol. 5, no. 11, Nov. 2011.

D. Lee, J. Choi, W. Lee, S. Pack, D.-Z. Du, and S. Hong. "A Time-Optimal Anti-collision Algorithm for FSA-Based RFID Systems," *ETRI Journal*, vol. 33, no. 3, pp. 458-461, Jun. 2011.

D. Lee, J. Choi, K. Kim, W. Lee, and C. Lee, "Optimal Transmission Policies for MOT in T-DMB," *IEEE Communications Letters*, vol. 11, no.12, pp. 1031-1033, Dec. 2007.

J. Choi, **D. Lee**, J. Yu, K. Kim, and W. Lee, "On the Calculation of the Maximal MOT Throughput in T-DMB," *IEEE Transactions on Consumer Electronics*, vol. 53, no. 3, pp. 877-884, Aug. 2007.

PEER-REVIEWED JOURNAL ARTICLES (DOMESTIC)

Y. Kim, D. Kim, **D. Lee**, J. Kim, M. Kwon, and M. K. Ahn, "Analysis of Mission Impact of Counter-Fire Operations due to Cyber Attacks using CyMIA," J. Internet Comput. Serv., vol. 25, no. 6, pp. 87–98, 2024.

Y. Kim, D. Kim, **D. Lee**, J. Kim, M. Kwon, and M. K. Ahn, "A Study on the Implementation of a Model for Mission Impact Assessment due to Cyber Attacks," J. Internet Comput. Serv., vol. 25, no. 5, pp. 1–11, 2024.

H. Kim, **D. Lee**, D. Kim, M. Ahn, and Y. Kim, "Policy Based DDoS Attack Mitigation Methodology," *Journal of KIISE*, vol. 43, no. 5, pp. 596-605, May 2016.

D. Lee, J. Park, C. Yoo, and H. Yun, "On the Design of a Big Data based Real-Time Network Traffic Analysis Platform," *Journal of the Korea Institute of Information Security & Cryptology*, vol. 23, no. 4, pp. 721-728, Aug. 2013.

INVITED JOURNAL ARTICLES (INTERNATIONAL/DOMESTIC)

이동환, 박찬일, 윤호상, "모바일 정보보호 동향 및 군적용 방안," 합동상호운용성, 제 7 호, pp. 78-89, 2012. 7.

INVITED BOOK-CHAPTERS

J. Choi, W. Lee, and **D. Lee**, "Comparative Performance Analysis of Anti-Collision Protocols in RFID Networks," *RFID Handbook: Applications, Technology, Security, and Privacy*, CRC Press, Boca Raton, FL, 2008.

CONFERENCE PROCEEDINGS (INTERNATIONAL)

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A. R. Thukkaraju, H. J. Yoon, S. Matsumoto, J. F. Ferrari, D. Lee, M. K. Ahn, "Interdependent Mission Impact Assessment of an IoT System with Hypergame-Theoretic Attack-Defense Behavior Modeling," in *Proc. 2023 31st IEEE Int. Symp. Modeling, Analysis, and Simulation of Computer and Telecommunication Systems (IEEE MASCOTS)*, 2023, pp. 1–8.

S. Matsumoto, J. F. Ferrari, H. J. Yoon, A. R. Thukkaraju, **D. Lee**, M. K. Ahn, et al., "Software-Friendly Subjective Bayesian Networks: Reasoning within a Software-Centric Mission Impact Assessment Framework," in *Proc. 2023 26th IEEE Int. Conf. Information Fusion (IEEE FUSION)*, 2023, pp. 1–8.

S. Lee, **D. Lee**, and S. Kim, "Block Double-Submission Attack: Block Withholding Can Be Self-Destructive," in *Proc. 4th ACM Conf. Advances in Financial Technologies (ACM AFT)*, 2023, pp. 232–243.

C. Park, **D. Lee**, H. Yun, and C. Kim, "Attribute-based Encryption Based on the Chain of Weighted Threshold Secret Sharing," In *Proceedings of Workshop on Computational Security*, pp. 1-5, Barcelona, Spain, Nov. 2011.

J. Choi, **D. Lee**, W. Lee, and D.-Z. Du, "RF-SWIFT: Beyond RF Identification and Probabilistic Collision Arbitration," in *Proceedings of IEEE International Conference on Wireless & Information Technology (IEEE CEWIT'2009)*, New York, USA, Oct. 2009.

D. Lee, K. Kim, and W. Lee, "*Q+-Algorithm*: An Enhanced RFID Tag Collision Arbitration Algorithm," In *Proceedings of the 4th International Conference on Ubiquitous Intelligence and*

Computing (UIC), Lecture Notes in Computer Science (LNCS), Vol. 4611, pp. 23 - 32, Hong Kong, China, 11-13 Jul. 2007. (Acceptance Ratio: 25.7% (119/463))

D. Lee, W. Lee, and J. Kim, "Genetic Algorithmic Topology Control for Two-Tiered Wireless Sensor Networks," In *Proceedings of the 7th International Conference on Computational Science (ICCS), Lecture Notes in Computer Science (LNCS)*, Vol. 4490, pp. 385-392, Beijing, China, 27 - 30 May 2007. (Acceptance Ratio: < 29.8% (716/2400))

J. Choi, J. Yu, **D. Lee**, S. Kim, and W. Lee, "Impacts of Repetitions and Segment Size Variability on Object-Level Throughput in T-DMB," In *Proceedings of 25th IEEE International Conference on Consumer Electronics (ICCE)*, Las Vegas, USA, 11-13 Jan. 2007.

W. Lee, E. Kim, J. Yu, **D. Lee**, J. Choi, J. Kim, and C. K. Shin, "UbiComm: An Adaptive Vertical Handoff Decision Scheme for Heterogeneous Wireless Networks," In *Proceedings of the 3rd International Symposium on Ubiquitous Computing Systems (UCS), Lecture Notes in Computer Science (LNCS)*, Vol. 4239, pp. 344 - 356, Seoul, Korea, 11 - 13 Oct. 2006. (Acceptance Ratio: 11.4% (41/359))

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Y. Kim, **D. Lee**, J. Kim, and M. K. Ahn, "A Study on Task Modeling Method for Executing Component-Level Models," in *Proceedings of 2024 KIMST Conference,* Jun. 2025.

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M. K. Ahn, D. Kim, **D. Lee**, J. Kim, Y. Kim, and M. Kwon, "Research on How to Configure and Utilize Ground-based Unmanned Vehicle Automated Resilience Diagnosis (GUARD)," In *Proceedings of 2024 KIMST Conference,* Jun. 2024.

Y. Kim, D. Kim, **D. Lee**, J. Kim, M. Kwon, and M. K. Ahn, "A Study on the Structural Design of Automated Penetration," In *Proceedings of 2024 KIMST Conference,* Jun. 2024.

W. Choi, D. Kim, J. Kim, Y. Kim, **D. Lee**, and M. K. Ahn, "Survey on CAN Bus-Off Attack," In *Proceedings of 2024 KIMST Conference,* Jun. 2024.

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D. Lee, D. Kim, M. K. Ahn, Y. Kim, J. Kim, and M. Kwon, "On the Design of LVC-Interoperable Gateway for Cyber Effect Emulation on UGV," In *Proceedings of 2025 KSII Spring Conference*, Apr. 2024.

Y. Kim, D. Kim, **D. Lee**, J. Kim, M. Kwon, and M. K. Ahn, "Implementation of Mission Damage Assessment Model due to Cyber Attacks," In *Proceedings of 2025 KSII Spring Conference*, Apr. 2024.

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J. Ahn, **D. Lee**, and M. K. Ahn, "A Research on Comprehensive Display for Trainee Monitoring on Cyberwarfare Training System," In *Proceedings of 2021 KSII Spring Conference*, Apr. 2021.

D. Lee, Y. Kim, and M. K. Ahn, "On Architecture Design for Collection of Tactical User Events in Cyber Training System," In *Proceedings of 2019 KIMST Conference,* Jun. 2019.

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H. Lee, **D. Lee**, C. Lee, Y. Kim, and M. K. Ahn, "A Study on Monitoring Techniques in Windows-based Cyber Training Environment," In *Proceedings of 2019 KIMST Conference,* Jun. 2019.

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C. Lim, B. Ko, and **D. Lee**, "Setting Priorities for Malware Distribution Sites using PageRank," In *Proceedings of KIISE Korea Computer Congress (KCC 2017)*, pp. 1057-1059, Jun. 2017.

J. Park and **D. Lee**, "Improving !exploitable with basic block expansion," In *Proceedings of KIISE Korea Computer Congress (KCC 2017)*, pp. 1066-1068, Jun. 2017.

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D. Lee, M. Ahn, and Y. Kim, "A Cyber Actuator System for Live-Virtual-Constructive Inter-Operative Cyberwarfare Simulation," In *Proceedings of 2016 KIISC Summer Conference*, pp. 359-362, Jun. 2016.

D. Kim, Y. Kim, J. Jim, W. Lee, **D. Lee**, and Y. Kim, "A Study of Conceptual Architecture for Cyber Range," In *Proceedings of 2015 Fall KIMST Conference*, pp. 791-792, Nov. 2015.

M. Ahn, **D. Lee**, D. Kim, and Y. Kim, "Research on Method of Cyber Attack Modeling in Constructive M&S Network Environment," In *Proceedings of 2015 Fall KIMST Conference*, pp. 773-774, Nov. 2015.

M. Ahn, **D. Lee**, H. Oh, and Y. Kim, "Research on MoE/MoP Effectiveness Analysis Based Large-Scale Networks," In *Proceedings of 2015 Fall KIMST Conference*, pp. 771-772, Nov. 2015.

D. Lee, D. Kim, M. Ahn, W. Cho, and Y. Kim, "On the Design of Distributed Agent for Emulating User Behavior & Traffic in Virtual Environment," In *Proceedings of 2015 Fall KIMST Conference*, pp. 855-856, Nov. 2015.

S. Lee, C. Lee, G. Kim, S. Nam, S. Shin, B. Roh, **D. Lee**, and Y. Kim, "사이버전 모의실험을 위한 네트워크 시뮬레이터 기반 모니터링 프레임워크," in *Proceedings of 2015 KINGComputing Fall Conference*, Nov. 2015.

B. Lee, S. Nam, S. Ihm, S. Shin, and B. Roh, **D. Lee**, and Y. Kim, "군 유선통신망에서 DDoS 공격 탐지 및 방어를 위한 IPSE 기반의 방화벽 모델 및 검증," in *Proceedings of 2015 KINGComputing Fall Conference*, Nov. 2015.

H, Kim, **D. Lee**, M. Ahn, Y. Kim, and K. Noh, "Measuring Technical Effectiveness in Cyber Warfare," In *Proceedings of KIISE Korea Computer Congress (KCC 2015)*, pp. 143-145, Jun. 2015.

H. Kim, **D. Lee**, M. Ahn, Y. Kim, E. Jung, and C. Pak, "Network Traffic Emulation for Cyber Range," In *Proceedings of KIISE Korea Computer Congress (KCC 2015)*, pp. 146-148, Jun. 2015.

M. Ahn, **D. Lee**, H. Oh, W. Cho, and Y. Kim, "Research on M&S System Architecture and Technology for Effect Analysis and Training/Test based Cyber Warfare," In *Proceedings of 2015 KIMST Conference*, pp. 1230-1231, Jun. 2015.

D. Lee, M. Ahn, Y. Kim, and K. Noh, "A Study on the Virtualization of Weapon Systems for Cyberwarfare Training, Test, and Evaluation," In *Proceedings of 2015 KIMST Conference*, pp. 1228-1229, Jun. 2015.

J. Jang, C. Park, **D. Lee**, H. Yun, "Security Architecture for Responding to Insider Cyber Threats," In *Proceedings of 2013 KIISC Summer Conference*, Jun. 2013.

D. Lee, C. Yoo, and H. Yun, "Big Data Analysis Based Malware Detection & Classification," In *Proceedings of 2013 KIMST Conference*, pp. 490-491, Jul. 2013.

PATENTS

INTERNATIONAL PATENTS

J. Park, K. Lee, S. Park, J. Woo, **D. Lee**, M. K. Ahn, "Ring-LWR-based Quantum-Resistant Signature Method and System Thereof," *US Patent No.: U 11,909,891 B2*, Feb. 20, 2024.

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DOMESTIC PATENTS

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D. Lee et al., "Method and Device for Authoring Integrated Scenarios for Mission Impact Analysis of Cyber War," *Korea Registration Number: 10-2812742*, May 2025.

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D. Lee et al., "Method for Constructing for Revocable Identity Based Encryption With Subset Difference Methods," *Korea Registration Number: 10-2348065*, Jan. 2022.

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D. Lee et al., "Apparatus and Method for Converting Real Threat Traffic Into Simulated Threat Traffic," *Korea Registration Number*: *10-2287926*, Aug. 2021.

D. Lee et al., "An Automatic Scores Allotment Method of Training Events," *Korea Registration Number*: *10-2242423*, Apr. 2021.

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